SYLLABUS

Embedded Intelligence at the Edge 7.5 credits D7065E

Inbyggd intelligens

Course syllabus admitted: Autumn 2026 Sp 1 - Present

DECISION DATE
2026-02-13

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Embedded Intelligence at the Edge 7.5 credits D7065E

Inbyggd intelligens

Second cycle, D7065E

Education level	Grade scale	Subject	Subject group (SCB)
Second cycle	G U 3 4 5	Datateknik	Computer Technology

Main field of study

Computer Science and Engineering

Entry requirements

At least 90 ECTS of completed courses and programming skills and basic knowledge in data communication, equivalent to the courses:
D0009E Introduction to Programming, 7.5 ECTS
D0010E Object-Oriented Programming and Design, 7.5 ECTS
D0002E Computer Communications, 7,5 ECTS
D0018E Database Technology, 7.5 ECTS
D0020E Project in Computer Engineering, 15.0 ECTS
D0029E Computer and Network Security, 7.5 ECTS
D7032E Software Engineering, 7.5 ECTS
Good knowledge of English, equivalent to English 6.

Selection

The selection is based on 30-285 credits

Course Aim

This course teaches how to design, build, and deploy intelligent edge solutions for cyber-physical systems (CPS) using Model-Based Systems Engineering (MBSE). You'll connect embedded systems, machine learning, and distributed computing to deliver secure, resilient, and efficient data-driven functions near the data source—across industrial IoT, autonomous systems, and smart environments.
After completing the course, students will be able to:

- Explain the design principles of edge intelligence and the edge–cloud continuum, including resource/data management and compute orchestration.
- Apply MBSE methods and tools for requirements analysis, architectural design, code generation, and system-behavior validation in distributed CPS, including data-analytics pipelines.
- Design, implement, and validate intelligent edge services in an industrial CPS setting, documenting results with model-based artifacts.
- Evaluate performance, latency, energy efficiency, and sustainability of edge-intelligence systems using MBSE.

Contents

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The course covers theory and practice in embedded intelligence, edge computing, and model-based CPS engineering, emphasizing industrially relevant solutions.
The focus areas are:

- Resource and data management and edge compute orchestration for distributed intelligence and autonomy in CPS.
- Model-Based Systems Engineering for CPS, including requirements management, functional decomposition, interface design, code generation, and validation.
- Edge computing solutions concerning security, virtualization, communication models, and data pipelines.

Realization

Each course occasion's language and form are stated and appear on the course page on Luleå University of Technology's website.
The language of instruction and the format of teaching are specified for each course occasion and can be found on the Luleå University of Technology website. The course is delivered as a distance course with the possibility to attend lectures and laboratory sessions in person. During the course, students participate in lectures and complete an individual assignment on model-based CPS engineering.

Examination

If there is a decision on special educational support, in accordance with the Guideline Student's rights and obligations at Luleå University of Technology, an adapted or alternative form of examination can be provided.
Acquired knowledge is demonstrated through a written report on MBSE modeling of intelligent edge services in an industrial CPS setting with oral examination, including the presentation and demonstration of the individual assignment.

Unauthorized aids during exams and assessments

If a student, by using unauthorized aids, tries to mislead during an exam or when a study performance is to be assessed, disciplinary measures may be taken. The term “unauthorized aids” refers to aids that the teacher has not previously specified as permissible aids and that may assist in solving the examination task. This means that all aids not specified as permissible are prohibited. The Swedish version has interpretative precedence in the event of a conflict.

Course offered by

Department of Computer Science, Electrical and Space Engineering

Modules

Code	Description	Grade scale	Cr	Status	From period	Title
0001	Report with oral exam and assignment	G U 3 4 5	7.5	Mandatory	A23	

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Study guidance

Study guidance for the course is to be found in our learning platform Canvas before the course starts. Students applying for single subject courses get more information in the Welcome letter. You will find the learning platform via My LTU.

Last revised

by Robert Brännström 2026-02-13

Syllabus established

by Robert Brännström, HUL at the Department of Computer Science, Electrical and Space Engineerin 2023-02-16

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